Lecture 7: Intro to computational methods

ECON 7510

Cornell University

Adam Harris

Motivation: Structural estimation

- Estimating structural models is a key element of the NEIO research process.
- These methods are potentially complementary to reduced-form methods.
  - Reduced form methods to establish facts that motivate the development of a structural model
  - Reduced form methods to establish a causal relationship; then,
    - structural model to decompose effects or mechanisms
    - structural model to evaluate counterfactual questions suggested by that relationship
  - Reduced form results can even be a direct input into strutural estimation (e.g., *indirect inference*)

Motivation: Structural estimation

- Estimating structural models is a key element of the NEIO research process.
- These methods are potentially complementary to reduced-form methods.
  - Reduced form methods to establish facts that motivate the development of a structural model
  - Reduced form methods to establish a causal relationship; then,
    - structural model to decompose effects or mechanisms
    - structural model to evaluate counterfactual questions suggested by that relationship
  - Reduced form results can even be a direct input into strutural estimation (e.g., *indirect inference*)
- How do we estimate model parameters?
  - Generalized method of moments / Simulated method of moments
  - Maximum likelihood estimator

Motivation: Structural estimation

- Estimating structural models is a key element of the NEIO research process.
- These methods are potentially complementary to reduced-form methods.
  - Reduced form methods to establish facts that motivate the development of a structural model
  - Reduced form methods to establish a causal relationship; then,
    - structural model to decompose effects or mechanisms
    - structural model to evaluate counterfactual questions suggested by that relationship
  - Reduced form results can even be a direct input into strutural estimation (e.g., *indirect inference*)
- How do we estimate model parameters?
  - Generalized method of moments / Simulated method of moments
  - Maximum likelihood estimator

→ **Question**: What do these methods have in common in terms of practical execution?

Motivation: Structural estimation

Extremum estimators

An *M-estmator* is characterized by

$$\hat{\theta} = \min_{\theta} \ell\left(x; \theta\right)$$

- For a simple model, might be able to solve FOCs for $\hat{\theta}$ analytically.
- For a more complex model, need to solve the optimization problem numerically.

→ In some cases, even the objective $\ell(x; \theta)$ might not be able to be expressed analytically.

  - Evaluating $\ell(x; \theta)$ might require integration → numerical integration
  - Evaluating $\ell(x; \theta)$ might require solving a nonlinear system of equations → iteration

- May not have analytical expression for SEs → bootstrapping

Overview of Topics

Theme: Fundamental tools of industrial organization

- **Part 1**: Fundamental IO theory
  1. Producer theory
  2. Monopolies: Pricing; product quality/choice
  3. Price discrimination
  4. Competition: Static; dynamic
- **Part 2**: Fundamental tools of structural estimation
  1. **Computational building blocks**
  2. Estimating demand systems
  3. Estimating models of single-agent dynamics
- **Part 3**: The IO research process

Overview of Topics

Computational building blocks

- **Today - 9/25**: Optimization methods
  - Gradient-based methods
  - Methods for constrained optimization
  - Methods for non-convex problems
- **9/25**: Numerical integration
  - Sampling methods
  - Quadrature methods
- **9/30**: Numerical methods in function spaces
  - Grids and interpolation
  - Solving fixed-point problems
  - Dynamic programming and value function iteration
- **10/2**: Non- and semi-parametric methods
  - Bias-variance tradeoff; overfitting
  - Mixture models; EM algorithm
  - Overview of ML methods: Tree-based models, NNs

Today’s goals

1. What programming language for what task?
2. Intro to Julia

Programming languages

My perspective

- **Stata**: My go-to for data cleaning, conventional reduced-form stuff
  - Pros: Great at what it's great at (merge, sort, by, expand, collapse)
  - Cons: Inflexible, clunky syntax; not the best tool for anything non-standard
- **R**: My go-to for some reduced-form stuff, ML, and visualization
  - Pros: Good packages and documentation.
  - Cons: Lacks Julia's speed
- **Julia**: My go-to for structural estimation
  - Pros: C-like performance (i.e., very fast)
  - Cons: Not as widely used; fewer and less well-maintained packages, less documentation
- **Matlab**: Improved a lot, but still way slower than Julia.
- **Python**: Tons of packages, very flexible and widely-used, but still slower than Julia.

Key Features of Julia

- **Performance:** Near C-level speed with Just-In-Time (JIT) compilation.
- **Multiple Dispatch:** Enables defining functions across different types.
- **Syntax:** Familiar to users of Python, Matlab, or R.
- **Rich Ecosystem:** Extensive libraries for scientific computing.
- **Open Source**

Julia Packages

Structure

- Base Julia comes with commands to do lots of basic operations.
- For other operations, need to import *packages*

Julia Packages

Structure

- Base Julia comes with commands to do lots of basic operations.
- For other operations, need to import *packages*

Potentially useful for structural estimation

- **Optim.jl:** Optimization routines.
- **DataFrames.jl**, **CSV.jl:** Data manipulation and analysis.
- **Econometrics.jl:** Econometric models and methods.
- **Random.jl, LinearAlgebra.jl, Distributions.jl**

Basics of Julia

See `Lec7.jl`